This brings us to the importance of the step and frequency responses. The *step response* describes how information represented in the *time domain* is being modified by the system. In contrast, the *frequency response* shows how information represented in the *frequency domain* is being changed. This distinction is absolutely critical in filter design because it is not possible to optimize a filter for both applications. Good performance in the time domain results in poor performance in the frequency domain, and vice versa. If you are designing a filter to remove noise from an EKG signal (information represented in the time domain), the step response is the important parameter, and the frequency response is of little concern. If your task is to design a digital filter for a hearing aid (with the information in the frequency domain), the frequency response is all important, while the step response doesn't matter. Now let's look at what makes a filter optimal for time domain or frequency domain applications.

Time Domain Parameters

It may not be obvious why the step response is of such concern in time domain filters. You may be wondering why the impulse response isn't the important parameter. The answer lies in the way that the human mind understands and processes information. Remember that the step, impulse and frequency responses all contain identical information, just in different arrangements. The step response is useful in time domain analysis because it matches the way humans view the information contained in the signals.

For example, suppose you are given a signal of some unknown origin and asked to analyze it. The first thing you will do is divide the signal into regions of similar characteristics. You can't stop from doing this; your mind will do it automatically. Some of the regions may be smooth; others may have large amplitude peaks; others may be noisy. This segmentation is accomplished by identifying the points that separate the regions. This is where the step function comes in. The step function is the purest way of representing a division between two dissimilar regions. It can mark when an event starts, or when an event ends. It tells you that whatever is on the *left* is somehow different from whatever is on the *right*. This is how the human mind views time domain information: a group of step functions dividing the information into regions of similar characteristics. The step response, in turn, is important because it describes how the dividing lines are being modified by the filter.

The step response parameters that are important in filter design are shown in Fig. 14-2. To distinguish events in a signal, the duration of the step response must be shorter than the spacing of the events. This dictates that the step response should be as *fast* (the DSP jargon) as possible. This is shown in Figs. (a) & (b). The most common way to specify the **risetime** (more jargon) is to quote the number of samples between the 10% and 90% amplitude levels. Why isn't a very fast risetime always possible? There are many reasons, noise reduction, inherent limitations of the data acquisition system, avoiding aliasing, etc.